Henry Liang

henry.liang1025@gmail.com henryliang-123.github.io 310-853-9937

Education

Northwestern University – M.S. in Machine Learning and Data Science Dec 2023

University of California, Los Angeles – B.S. in Applied Mathematics, Statistics Mar 2022

Work Experience

Senior AI Engineer, Vultron AI – San Francisco, CA Apr 2025 – Present

- Led the redesign of the LLM integration layer to future-proof the system for multi-provider support, reducing model adoption time from weeks to days
- Oversaw AI infrastructure migration and created multiple AI agents to autonomously write, execute, and verify prompt regression tests and BAML files, accelerating project timeline from 3 months to 3 weeks
- Improved Human-AI interactions by designing a collaborative agentic proposal assistant and fine-tuning LLMs to use customer-specific language and content
- Coordinated on-call team to reduce CUDA OOM errors by 99% for customer deployments by redesigning OCR and reranker services using model pools, custom configurations, and improved observability
- Designed and integrated an AI evaluation system to establish quantitative baselines and allow continuous monitoring for retrieval accuracy, hallucination levels, and general output quality

AI Engineer, Vail Systems, Inc. – Chicago, IL Feb 2024 – Apr 2025

- Led the development and fine-tuning of a Retrieval-Augmented Generation (RAG) system, achieving 99.18% top-5 retrieval accuracy to enhance enterprise question-answering capabilities.
- Spearheaded the engineering of an autonomous AI agent that analyzes daily call transcripts, extracts key insights, and generates executive summaries on call trends, sentiment drivers, and performance metrics
- Created a novel hybrid embedding that reduces the no-context rate for request-for-proposal completion by 83.33%, leading to a paper acceptance and invited talk at AI-ML Systems Conference, October 2024

Applied Scientist Intern, Amazon (Capstone) – Evanston, IL Sept 2023 – Dec 2023

- Designed an LLM application to perform graph reasoning for network routing anomaly detection, achieving 100% accuracy on core graph problems and scaling to graphs with 10K+ nodes
- Reduced erroneous graph database queries by 95% via prompt engineering and continuous fine-tuning of the query generation process using a few-shot learning feedback loop
- Co-authored paper on Graph Reasoning via Retrieval Augmented Framework (GRRAF), accepted for publication at EMNLP 2025

Machine Learning Researcher, Northwestern University – Evanston, IL Oct 2022 – June 2023

- Led design and implementation of a scalable AWS Internet of Things (IoT) streaming pipeline for real-time time-series forecasting; achieved a throughput of up to 55 predictions per second
- Trained and deployed an LSTM model on AWS EC2 for hourly Divvy bike demand forecasting for each station; realized a mean absolute error of 9 bikes

Skills

Languages and Tools: Python, R, PostgreSQL, Java, JavaScript, Rust

Tools: TensorFlow, PyTorch, LangChain, Hugging Face, Git, Docker, Kubernetes, Apache Spark, Apache Hadoop